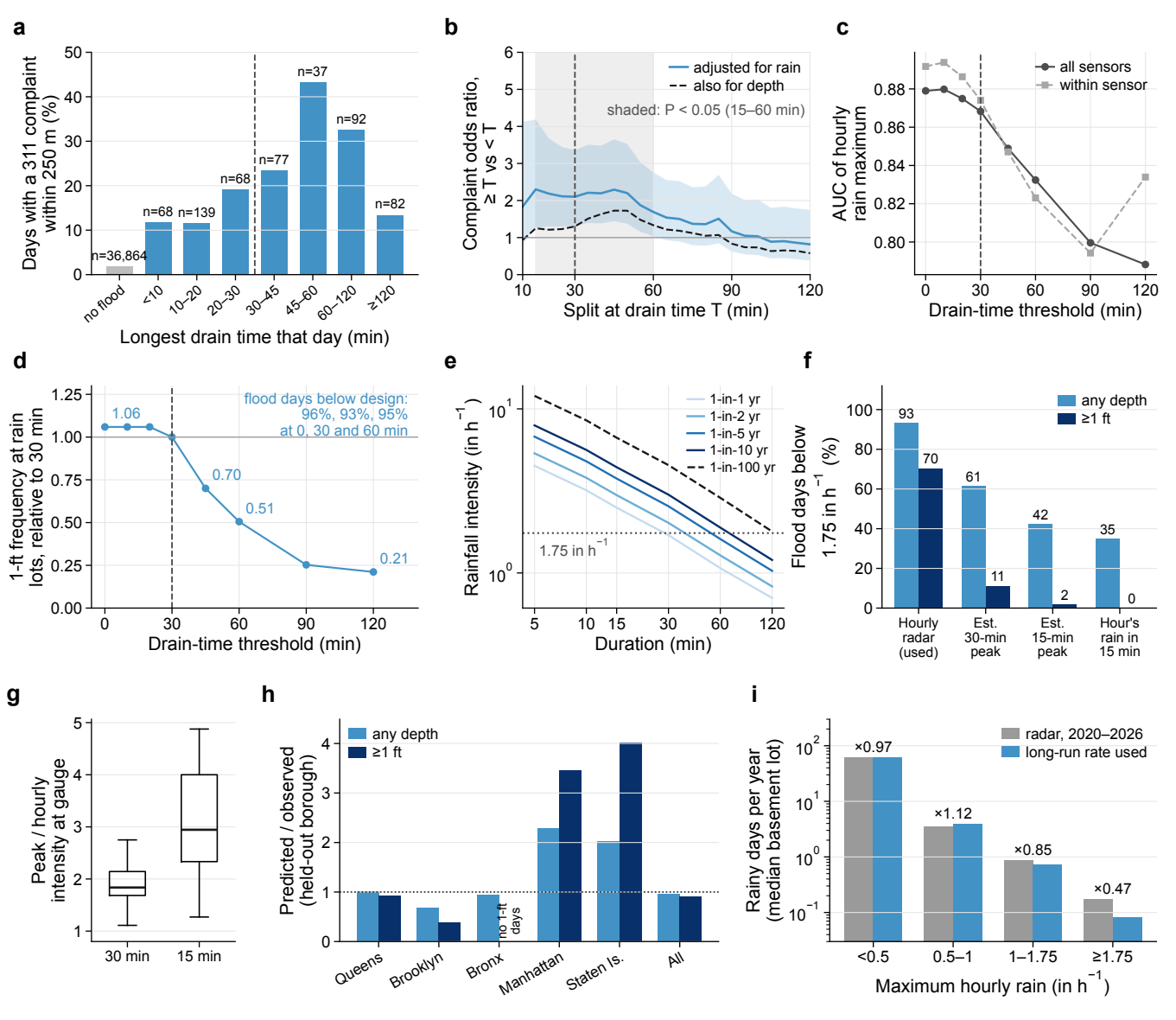
a
Days with a 311 complaint within 250 m (%)
n=36,864
n=68
n=139
n=68
n=77
n=37
n=92
n=82
no flood
<10
10–20
20–30
30–45
45–60
60–120
≥120
Longest drain time that day (min)
b
Complaint odds ratio, ≥ T vs < T
adjusted for rain
also for depth
shaded: P < 0.05 (15–60 min)
Split at drain time T (min)
c
AUC of hourly rain maximum
all sensors
within sensor
Drain-time threshold (min)
d
1-ft frequency at rain lots, relative to 30 min
flood days below design:
96%, 93%, 95%
at 0, 30 and 60 min
1.06
0.70
0.51
0.21
Drain-time threshold (min)
e
Rainfall intensity (in h⁻¹)
1-in-1 yr
1-in-2 yr
1-in-5 yr
1-in-10 yr
1-in-100 yr
1.75 in h⁻¹
Duration (min)
f
Flood days below 1.75 in h⁻¹ (%)
any depth
≥1 ft
93
70
61
11
42
2
35
0
Hourly radar (used)
Est. 30-min peak
Est. 15-min peak
Hour's rain in 15 min
g
Peak / hourly intensity at gauge
30 min
15 min
h
Predicted / observed (held-out borough)
any depth
≥1 ft
no 1-ft days
Queens
Brooklyn
Bronx
Manhattan
Staten Is.
All
i
Rainy days per year (median basement lot)
radar, 2020–2026
long-run rate used
×0.97
×1.12
×0.85
×0.47
<0.5
0.5–1
1–1.75
≥1.75
Maximum hourly rain (in h⁻¹)